

Trust and Resilience in Connected Aviation Systems

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Abstract— Connected aviation systems increasingly depend on external inputs that are treated as authoritative by default. Using two documented incidents—a 2022 analysis of Global Navigation Satellite Systems (GNSS) disruptions in European airspace based on Automatic Dependent Surveillance–Broadcast (ADS-B) observations and the 2025 hacktivist distributed denial of service (DDoS) attack against Los Angeles International Airport’s public-facing services—this paper examines how implicit trust in positioning, navigation, and timing (PNT) accuracy and in service availability allows erroneous or missing data to influence operations before monitoring alarms converge, enabling cascades. A cross-case, socio-technical analysis identifies where trust assumptions are embedded in workflows and derives resilience design directions. Findings indicate that technical safeguards are necessary but insufficient. Impact is bounded most effectively when trust calibration is paired with resilience-by-design mechanisms. The study proposes operational metrics—mean time to detect (MTTD), mean time to recover (MTTR), recovery time objectives, and cascade depth (CD)—to evaluate improvements across incidents and exercises.

Keywords— aviation cybersecurity, critical infrastructure, implicit trust, resilience engineering, GNSS disruption, denial-of-service, degraded-mode operations

I. INTRODUCTION

Aviation has emerged as one of the most critical infrastructures globally, with GNSS serving as the backbone for modern air traffic management, navigation, and surveillance operations [1], [2]. The transition toward Performance-Based Navigation (PBN) and space-based air navigation has significantly increased aviation’s reliance on GNSS technologies, making these systems indispensable for safe and efficient flight operations [1], [3]. However, this transformation has introduced a complex web of implicit trust relationships that permeate every level of aviation operations. From pilots’ trust in automated navigation systems to air traffic controllers’ reliance on GNSS-derived aircraft position data, these trust relationships have become fundamental to aviation safety and efficiency [4]. Yet, they also represent potential vulnerabilities that can be exploited by malicious actors or lead to catastrophic failures when trust assumptions prove incorrect. The period from 2020-2025 has witnessed an alarming escalation in GNSS interference incidents, with sophisticated attacks becoming more frequent and widespread [5]. These vulnerabilities are particularly concerning given that GNSS signals are inherently weak when received on Earth’s surface, making them highly susceptible to both unintentional and intentional interference [2]. Critical challenges persist in the implicit trust placed on GNSS signals and integrated navigation systems, leading to the elimination or reduction of traditional manual verification procedures [6], [7]. This shift has introduced vulnerabilities, as the aviation community increasingly relies on automated fault detection and exclusion mechanisms without

independent position determination [8], [9]. Several studies highlight the challenges posed by urban environments, cyber threats, and satellite faults that can degrade navigation integrity [8], [10], [11]. Moreover, controversies exist regarding the balance between automation benefits and the potential for pilot complacency and skill degradation, which may exacerbate risks during system failures [6], [7].

The consequences of these gaps include increased integrity risks, reduced continuity of navigation services, and potential safety hazards in critical flight phases [12], [13]. This study is guided by the following research question: Which resilience capabilities most effectively reduce operational impacts during GNSS interference and DDoS disruptions in civil aviation, and how does trust calibration influence their effectiveness? To address this question, the paper sets three objectives. First, it characterizes GNSS spoofing as a representative cyber-physical threat in aviation, framing it as a failure of implicit trust rather than a signal-level anomaly. Second, it evaluates resilience-oriented design strategies—detection, mitigation, and recovery—in the context of GNSS interference and DDoS events, assessing their role in reducing operational disruption. Third, it examines trust calibration practices as mechanisms for balancing operator confidence and system autonomy under degraded navigation conditions.

II. BACKGROUND AND RELATED WORK

A. Aviation as Critical Infrastructure

Research on the role of GNSS in aviation has emerged as a critical area of inquiry due to its fundamental impact on PNT services essential for safe and efficient air traffic management (ATM) and airline operations [14], [15]. Since the introduction of satellite-based navigation, aviation has transitioned from reliance on ground-based aids to GNSS-enabled PBN, enhancing route flexibility and precision [14], [16]. The evolution of GNSS, including multi-constellation and multi-frequency systems such as Global Positioning System (GPS) and Galileo, has improved navigation accuracy and integrity, supporting advanced approach and landing procedures [17], [18]. Despite these advances, the increasing dependence on GNSS exposes aviation to vulnerabilities from radio frequency interference (RFI), jamming, and spoofing, which can degrade or disrupt satellite signals [19], [20]. Given aviation’s status as critical infrastructure, GNSS disruptions pose significant risks to safety, economic stability, and national security, underscoring the importance of resilient navigation solutions [15], [21]. Current literature reveals a knowledge gap in comprehensive understanding and mitigation of GNSS interference impacts on ATM and aircraft operations, particularly under adverse conditions such as space weather or intentional interference [1], [22], [23]. Below, we outline key differences between pre-2022 and 2022–2025 GNSS disruption patterns—covering scale, attack modalities, and operations—with citations provided in the narrative. A cross-wave comparison shows GNSS interference has shifted markedly since 2022. Scale has expanded from localized, site-specific hotspots to regional corridor-

level interference spanning hundreds to thousands of km² [24], [25]; persistence has moved from short, episodic events to recurrent, prolonged campaigns lasting days to weeks and sometimes months [26]. Spoofing has matured from single-system, single-frequency attacks with noticeable jumps to multi-constellation, multi-frequency, trajectory-coherent carry-offs [27], while attack vectors have broadened beyond GPS L1 to coordinated spoofing/jamming across GPS, GLONASS, Galileo, and BeiDou, including L1/L5 [27], [28]. Aviation impacts have intensified, from minimal or rare commercial effects to documented Required Navigation Performance (RNP) degradation/loss, reroutes, go-arounds, and capacity reductions [5], [29]. Correspondingly, detection and mitigation have evolved from Receiver Autonomous Integrity Monitoring (RAIM) checks, pilot reports, and localized Notice to Air Missions (NOTAMs) to sensor/data fusion (GNSS + INS/DME/baro), anti-jam antennas, fleet-wide and Air Navigation Service Provider (ANSP) monitoring, and collaborative advisories [30], [31]. Operational posture has shifted from ad hoc advisories to coordinated mitigation with updated SOPs, multinational information sharing, and dedicated training modules, tactical checklists, and preemptive planning for jamming/spoofing [32], [33].

B. Jamming

Research has identified distinct geographical patterns in GNSS interference activities, with Eastern Europe and the Middle East emerging as primary hotspots for both jamming and spoofing attacks. A comprehensive analysis of air traffic data from February to August 2022 revealed widespread GNSS jamming across Eastern Europe, with indiscriminate interference from Russian electronic warfare units, emanating from the Russian mainland, Kaliningrad, and conflict zones abroad, posing serious hazards to aircraft navigation and operations [19], [34]. Quantitative analysis demonstrates the scale of impact: researchers observed several days when more than 1,000 flights were affected, representing 60% of the daily traffic in the analyzed Eastern European airspace [35]. The interference exhibits varying strength and duration, with altitude-dependent effects where low-flying aircraft tend to be less affected than those operating at higher altitudes [35]. The manipulation or jamming of GNSS signals in this region severely compromises not only aviation operations but also weather balloon tracking networks, leading to inaccurate atmospheric measurements and undermining weather forecasting accuracy [36]. The practical significance is underscored by recent statistics indicating thousands of daily affected flights in Europe alone and widespread jamming events near airports and highways [2], [37].

C. Spoofing

The Middle East region presents equally concerning patterns of GNSS interference, with comprehensive analysis using software-defined radio (SDR) systems revealing detailed findings on the prevalence and characteristics of satellite navigation system spoofing attacks [36]. However, the scale and impact of the current wave of spoofing are unprecedented. In the initial months, only a small number of aircraft were affected. By January 2024, an average of 300 flights each day experienced spoofing incidents. This figure surged to approximately 1,500 flights per day by August 2024. Most recently, between July 15 and August 15, 2024, a total of 41,000 flights encountered GPS spoofing [25]. Regulatory bodies such as the FAA and EASA have issued warnings highlighting the growing threat from GPS spoofing to aviation safety and operational integrity [38], [39]. These developments emphasize the urgent need to understand and mitigate GPS spoofing to safeguard safety-of-life applications [25], [40], [41].

D. Problem statement and research gap

The specific problem addressed is the increasing frequency and sophistication of GNSS spoofing and jamming attacks, which

compromise the integrity and availability of satellite navigation signals [34], [42]. Despite advances in detection technologies and a range of proposed solutions, substantial gaps persist in understanding the geographic spread, operational impact, and efficacy of countermeasures—underscoring the urgent need for real-time, broadly adaptable detection techniques capable of addressing varied attack scenarios and platforms [43], [44], [45], [46], [47]. Ongoing debates center on the trade-offs between hardware-dependent methods, which deliver high accuracy but entail greater cost and system complexity, and software-based machine-learning approaches, which offer scalability yet may be prone to false alarms or demand extensive training datasets [48], [49], [50]. Disagreement also surrounds the predictability of interference events: while some studies link disruptions primarily to geopolitical tensions, others highlight unintentional or criminal origins [34], [51]. The consequences of these gaps include potential aviation accidents, compromised fleet operations, and degraded critical infrastructure performance [52], [53]. Addressing these uncertainties is essential to enhance resilience and inform policy and technical responses [38], [54]. Progress on detection, authentication, and redundancy has been substantial, yet the literature remains dominated by technical countermeasures.

What is missing is the socio-technical layer: the assumptions and trust relationships baked into system design and operations. Today's aviation ecosystems treat GNSS-derived PNT as authoritative by default—signals are ingested by avionics, flight management, Air Traffic Control (ATC) decision-support, and dispatch tools with limited, often procedural, cross-validation. When an adversary induces spoofing or sustained jamming, this unexamined, implicit trust allows bad inputs to propagate across navigation, separation, and scheduling, creating cascading operational effects rather than isolated faults. Although resilience engineering emphasizes anticipation, graceful degradation, rapid recovery, and adaptation under stress; in aviation, however, practice still skews toward component-level redundancy (e.g., more sensors) rather than system-level resilience (e.g., multi-sensor “truthing,” anomaly triage, mode transitions with bounded performance, recovery time objectives for PNT loss, crew/ATC playbooks, and post-event learning loops). As a result, the interaction between trust calibration and resilience mechanisms remains under-theorized and under-measured. This paper addresses that gap by treating GNSS interference as a failure of implicit trust and examining how targeted resilience strategies can bound cascade risk and sustain safe operations.

E. Cyber Threats in Aviation

ADS-B vulnerabilities are the most consistently documented aviation cyber threat in the literature. Because ADS-B is an open, unencrypted, and unauthenticated broadcast, receivers cannot cryptographically verify message origin or integrity, enabling data injection, modification, replay, and spoofing [55], [56]. The widespread availability of low-cost SDRs and small unmanned aircraft further lowers the barrier to transmitting convincing but fraudulent ADS-B messages, degrading situational awareness [55]. Beyond ADS-B, airport information-technology infrastructure has become a high-value ransomware target as digitization expands the attack surface [57]. Supply-chain risk is also prominent, with multiple studies identifying firmware integrity as an “Achilles’ heel” for aviation cyber-physical systems [58], [59].

III. RESILIENCE FRAMEWORK

Aviation systems traditionally operate under strong implicit trust assumptions, where components assume the authenticity and reliability of data from other system elements [60], [61]. This trust model, while enabling efficient operations, creates systemic vulnerabilities when adversaries can exploit trusted communication channels. Research identifies concerning patterns of “trust decay” where security measures degrade over time due to operational pressures and cost considerations

[62]. Blind trust in vendor-provided components constitutes a distinct systemic risk, and excessive reliance on proprietary architectures undermines the capacity for independent security evaluation and resilience assurance. Traditional aviation resilience focuses primarily on physical redundancy and fail-safe mechanisms [63]. However, cyber threats require different resilience strategies that account for the interconnected nature of digital systems and the potential for cascading failures. A significant research gap exists in standardized metrics for measuring cyber-resilience in aviation systems [64]. Current approaches focus primarily on availability metrics rather than comprehensive resilience assessment.

To systematically address these identified gaps, aviation cybersecurity resilience capabilities must be aligned with established governance frameworks and regulatory standards. A comparative mapping to NIST CSF 2.0 [65], Govern outcomes, NIST SP 800-82 [66], operational technology (OT) constraints, and RTCA DO-326B [67]/DO-356B [68] airworthiness security activities offers a structured analytical lens through which strengths can be evaluated and critical deficiencies more precisely identified. Theoretical alignment among a high-level governance taxonomy, OT operational constraints, and aviation airworthiness standards is essential to achieve consistent governance, but each source emphasizes different objectives and granularity, which complicates direct mapping and accountability relationships. The CSF 2.0 Govern function defines high-level governance outcomes and links to guidance rather than prescriptive controls, while OT literature emphasizes legacy device constraints and real-time availability, and RTCA DO-326B/DO-356B impose rigorous lifecycle and airworthiness evidence requirements that are more prescriptive for airborne systems [65], [69], [70]. To provide a structured comparison, (A. Comparative Attributes and Integration Friction points) contrasts the primary aims, prescriptiveness, lifecycle coverage, deliverables, and integration challenges across NIST CSF 2.0 (Govern), NIST SP 800-82 OT constraints, and RTCA DO-326B/DO-356B aviation standards. This comparison demonstrates that while each framework addresses resilience, their differences in scope and enforceability complicate alignment

A. Comparative Attributes and Integration Friction Points

Primary Aim

- NIST CSF 2.0 (Govern): High-level governance outcomes, risk communication, and policy alignment [65].
- NIST SP 800-82 OT constraints: Protect availability and integrity of control loops; tolerate legacy and deterministic timing [70]
- RTCA DO-326B/DO-356B aviation: Airworthiness-grade security risk assessment, lifecycle evidence, and certification artifacts [69].

Prescriptiveness

- NIST CSF 2.0 (Govern): Non-prescriptive; maps to many control sources [65].
- NIST SP 800-82 OT constraints: Technical limitations force pragmatic mitigations (segmentation, micro-controls) [70].
- RTCA DO-326B/DO-356B aviation: Prescriptive SRA/TARA processes and artifact outputs to satisfy certification [69].

Lifecycle Coverage

- NIST CSF 2.0 (Govern): Governance across design, operations, supply chain, and continuous improvement [65].
- NIST SP 800-82 OT constraints: Focus on operational continuity, patching challenges, and maintenance windows [70].
- RTCA DO-326B/DO-356B aviation: Strong emphasis on design-time SRA and continued airworthiness evidence [69].

Typical Governance Deliverable

- NIST CSF 2.0 (Govern): Risk tolerance statements, roles, policy, and supplier oversight guidance [65].
- NIST SP 800-82 OT constraints: Network zoning, safe-state controls, and constrained patching policies [70].

- RTCA DO-326B/DO-356B aviation: Security Risk Assessment (SRA), Threat Assessment and Remediation Analysis (TARA), and traceable mitigations [69].

Key Integration Challenge

- NIST CSF 2.0 (Govern): Translating high-level risk tolerance into certifiable avionics artifacts and OT operational rules [65], [69].
- NIST SP 800-82 OT constraints: Mapping prescriptive avionics artifacts to OT practices limited by legacy and real-time constraints [69], [70].
- RTCA DO-326B/DO-356B aviation: Ensuring governance assigns accountability for procurement and supplier evidence so airworthiness artifacts are auditable and implementable [65], [69].

To advance the study's goal of identifying how implicit trust and resilience challenges manifest within aviation cybersecurity, it is not sufficient to map frameworks in isolation; it is equally important to examine the structural misalignments that emerge when governance, OT operational constraints, and certification requirements intersect. Cross-framework analyses (as shown in Table I) reveal substantive areas of misalignment.

TABLE I. CROSS-FRAMEWORK MISALIGNMENTS IN AVIATION CYBERSECURITY RESILIENCE

Gap	NIST CSF 2.0 (Govern)	NIST SP 800-82 (OT Constraints)	RTCA DO-326B/DO-356B (Airworthiness Security)
Scope mismatch	Defines broad governance outcomes and risk tolerances [65], [69].	Provides OT security principles but not certification artefacts [66].	Requires specific SRA/TARA artefacts with traceability for certification, creating a gap when translating broad policy into certifiable evidence [69].
Operational feasibility	Envisions risk-based mitigations at governance level [70].	Legacy devices, deterministic timing, and safety-critical constraints limit feasible mitigations [66].	Certification assumes evidence of effective mitigation, but feasibility may be constrained by OT realities [71].
Evidence & accountability	Calls for governance, accountability, and supplier oversight [65], [71].	Limited guidance on contractual evidence flows in OT contexts [66].	Requires lifecycle evidence, supplier artefacts, and traceability; governance often fails to mandate robust evidence flows [71].

First, a scope incongruity is evident: whereas the NIST Cybersecurity Framework (CSF) "Govern" function articulates broad governance outcomes and risk tolerance principles, RTCA DO-326B and DO-356B prescribe the production of specific, certifiable artefacts—such as Security Risk Assessments (SRA) and Threat Assessment and Remediation Analyses (TARA)—with demonstrable traceability for certification purposes [65], [69]. This divergence generates a translation gap between high-level policy intent and the evidentiary requirements of regulatory certification. Second, operational feasibility is constrained by the deterministic, safety-critical characteristics of legacy operational technology (OT) environments, as delineated in NIST SP 800-82, which limit the applicability of governance prescriptions framed at a strategic level

[70]. Finally, aviation standards place pronounced emphasis on lifecycle evidence and supplier-generated artefacts; consequently, governance functions must explicitly define oversight mechanisms and contractual evidence-flow requirements—elements that, in practice, are frequently underdeveloped or inconsistently enforced [65], [69].

IV. METHODOLOGY / ANALYTICAL APPROACH

This study adopts a qualitative multiple-case study design to examine how trust assumptions and resilience practices shape security outcomes in connected aviation systems. Each case is analyzed independently (within-case) and then compared (cross-case) using replication logic to strengthen analytic generalization. Limited descriptive quantification (e.g., durations, shares, time-to-recover) supports explanation, but the emphasis remains qualitative.

A. Units Case Selection and Boundaries

We purposively selected two information-rich cases that contrast operational tiers (airborne vs. ground) and threat modalities (signal disruption vs. enterprise IT disruption):

Case 1 — GNSS Disruptions (Airborne Navigation): Analysis of GNSS Disruptions in European Airspace (Open Sky ADS-B, Feb–Aug 2022) forms the empirical basis. Spatial boundaries are the three Areas of Interest (Baltics/Kaliningrad; Romania/Black Sea; Cyprus/Eastern Mediterranean) [37]. The dataset highlights navigation anomalies across multiple FIRs, exhibiting signatures that align with known interference and disruption patterns. While attribution is an ongoing process, the consistency of these anomalies across regions indicates systemic stress on navigation reliability rather than isolated technical faults.

Case 2 — LAX Hacktivist DDoS (Ground Infrastructure): The March 18, 2025, DDoS attack on Los Angeles International Airport’s public-facing services is documented via a peer-reviewed synthesis of aviation cyber risks and sector reporting. The incident boundary spans initial impact through service stabilization, focusing on FIDS/portal/e-check-in availability and operational fallbacks. [33], [72]. In this case, we employ triangulation of peer-reviewed scholarship on aviation cybersecurity risks with sectoral incident reporting to reconstruct the impacts of disruption events and the corresponding organizational response patterns.

B. Unit of Analysis

Case 1 (GNSS disruptions): Flight-segment traversing a predefined RFI Area of Interest (Aoi)—the contiguous interval from AoI entry to AoI exit for a given flight.

Case 2 (LAX DDoS): Airport-incident affecting public-facing digital services—the event window covering initial impairment, mitigations, and restoration of passenger-facing IT (web/app/info systems).

C. Data Sources and Collection

Case 1:

- Source & period. Crowd-sourced, public ADS-B from the OpenSky Network, covering Feb 1–Aug 31, 2022; 1,034,455 flights processed [37].
- Tables & merge. Positions from `state_vectors_data4`; ADS-B status (for NACp) decoded from `operational_status_data4`; datasets merged by `icao24` and timestamp (processed with the traffic library) [37].
- Areas of Interest (AoIs). Three European AoIs named by EASA and cross-checked via GPSJam were selected for detailed analysis [37].

Unit handled downstream: portions of flights inside AoIs are the basis for subsequent labeling and summaries (e.g., affected-flight share, durations) [37].

Case 2:

(LAX DDoS secondary synthesis). Publicly available incident documentation—anchored by SOCRadar’s report on the Mar 18, 2025 hacktivist DDoS at LAX—and peer-reviewed aviation-cybersecurity literature (to contextualize airport service criticality and common IT failure points) are synthesized to reconstruct the event timeline and service impairments and to derive time-to-restore (TTR) and % of services operated in degraded mode, with estimates tagged by source and confidence [72]. Sector reporting on 2025 aviation cyber incidents with details on attack type (volumetric/multi-vector DDoS), claimed attribution (hacktivist), and operational impacts on portals/FIDS/check-in services [33].

D. Construct and Operationalization

- Trust (implicit/operational).
 - Case 1 proxies: implicit trust in unauthenticated GNSS signals; single-signal reliance for PBN; limited cross-checks at the signal level.
 - Case 2 proxies: availability trust in public portals/FIDS/kiosk backends; reliance on vendor-managed WAF/scrubbing/CDN capacity without adversarial surge validation.
- Resilience (anticipate–absorb–recover–adapt).
 - Anticipate: threat modeling, monitoring, exercises (RFI watchlists; DDoS intel and capacity tests).
 - Absorb: ability to sustain operations during disruption (sensor-fusion and nav aids; cached FIDS, manual check-in, PA/SMS messaging).
 - Recover: time-to-recover (TTR), restoration sequencing, backlog clearance.
 - Adapt: post-incident hardening (receiver robustness, multi-constellation; anycast + on-demand scrubbing, vendor SLA stress tests).

E. Analytic Strategy

- Within-case analysis uses a simple timeline (onset → detection → containment → recovery), pattern-matching to trust/resilience constructs, and matrix tables. Metrics are TTR and % of services degraded. For LAX (Mar 18, 2025): TTR = not determined (insufficient public evidence); ~60% degraded (low confidence; SOCRadar-based) [33].
- Cross-case synthesis: a meta-matrix compares trust assumptions, failure modes, resilience capabilities, and outcomes across cases; explanation building identifies common mechanisms (e.g., single-layer dependencies) and derives design principles for trust-aware, resilience-engineered aviation systems.

F. Rigor: Reliability and Validity

- Construct validity. We strengthened construct validity by triangulating across data types (e.g., ADS-B observations, advisories/NOTAMs, operator reports), using an explicit codebook to operationalize “trust” and “resilience,” and maintaining an audit trail of coding and inclusion decisions.
- Internal validity. We examined rival explanations for each finding—for example, distinguishing radio-frequency interference (RFI) from coverage artifacts in ADS-B data, and separating IT misconfiguration from deliberate DDoS through timing, traffic characteristics, and provider confirmations.
- External validity. The goal is analytic generalization to theory, trust, and resilience in connected critical infrastructure, rather than statistical generalization; we state boundary conditions under which the insights are expected to hold.
- Reliability. We used standardized extraction templates, reproducible ADS-B parsing steps, and a documented case

database; a sample of cases was double-coded to assess consistency, with disagreements resolved by consensus and logged.

To preserve provenance separation, Case 1 relies solely on primary operational traces from OpenSky: we follow the published ADS-B-based interference method (tracking NACp degradation and recovery within defined Areas of Interest) to label affected flight segments and then summarize affected-flight share and exposure durations; no RAIM/FMS states are inferred from ADS-B [37]. Case 2 instead compiles secondary documentation of the 18 Mar 2025 LAX hacktivist DDoS, airport/airline communications, regulator advisories, and peer-reviewed sector reporting, from which we extract the event timeline and service impacts to derive TTR and degraded-services percentages [33]. Results are reported separately and not pooled, reflecting the distinct evidence types (airborne navigation telemetry vs. ground-service incident records)

V. CASE STUDIES & DISCUSSION

A. Case 1- GNSS Disruptions in European Airspace

This case synthesizes findings from [39] which analyzed GNSS disruptions in European airspace using OpenSky Network ADS-B data (Feb.–Aug. 2022), as summarized in Table II. The study identified three areas of interest—Baltics/Kaliningrad, Romania/Black Sea, and Cyprus/Eastern Mediterranean—and detected affected flights via navigation accuracy and integrity degradation (e.g., NACp shifts). Reported operational impacts included increased pilot workload, reliance on ATC vectors, and variability in resilience depending on aircraft equipage (IRS/DME versus GNSS-only).

TABLE II. KEY FINDINGS — FLIGHT DISRUPTIONS

AoI	Flights Analyzed	% Affected	Hotspots
1. Baltics–Kaliningrad	511,195	1.60%	Kaliningrad, S. Lithuania, NE Poland [37]
2. Romania–Black Sea	428,118	14%	Black Sea corridor, Moldova [37]
3. Cyprus–E. Mediterranean	95,142	70%+	S. Turkey, Syria, Lebanon [37].

Trust implications: implicit reliance on unauthenticated GNSS signals; limited cross-checking. Resilience implications: need for hardened receivers, sensor fusion (IRS/DME/multi-constellation), and preservation of ground-based nav aids.

B. Case 2 – LAX Hacktivist DDoS (March 18, 2025)

Case Summary

On March 18, 2025, Los Angeles International Airport (LAX) experienced a large-scale distributed denial-of-service (DDoS) attack publicly claimed by a hacktivist collective. Reported impacts included degraded availability of public-facing portals, disruptions to flight information displays (FIDS), and intermittent issues with online/electronic check-in. Core flight operations remained safe; staff activated manual procedures for passenger services.

Technical Characteristics

- Volumetric attacks against Internet-facing web properties and FIDS integrations, leveraging botnet-scale traffic.
- Multi-vector execution across volumetric, protocol, and application layers.
- Coordination with social media narratives to amplify perceived disruption.

- Effects concentrated on availability; integrity/confidentiality of core systems not reported as compromised.

Attack Execution Patterns

- Exploitation of insufficient DDoS mitigation capacity at the network perimeter and upstream providers.
- Stress on content delivery and API backends feeding terminal displays and self-service check-in.
- Temporary denial for passengers using web and kiosk channels; manual fallbacks engaged.

Trust Analysis (Implicit & Operational)

- Availability Trust: Operations assume continuous reachability of portals, FIDS APIs, and kiosk backends. DDoS breaks this assumption—data remain intact but inaccessible—exposing blind trust in network reachability and vendor/CDN capacity.
 - Vendor/Process Trust: Reliance on third-party scrubbing/WAF services without validated surge capacity under adversarial load.
- #### Resilience Analysis (Anticipate–Absorb–Recover–Adapt)
- Anticipate: Heightened hacktivist DDoS activity in 2025 made such attacks foreseeable; capacity tests and playbooks required.
 - Absorb: Limited absorption—public websites and FIDS degraded; manual wayfinding and check-in required; flight safety unaffected.
 - Recover: Services restored as mitigations engaged, and hostile traffic subsided; no lasting damage reported.
 - Adapt: Anycast + on-demand scrubbing with contracted burst capacity; signed, cached ‘degraded mode’ FIDS; decoupled passenger messaging (SMS/PA); rehearsed runbooks for manual check-in and queue control.

C. Broader Ground-Side Threat Context (Ransomware/Extortion)

Beyond DDoS, 2025 aviation incidents show ransomware/extortion campaigns that degrade passenger-facing services via encryption + exfiltration (double extortion), commonly entering through phishing/exposed remote access, spreading via weak segmentation, and timing deployment for weekends/holidays; several airports reported >10 h of manual operations and backup targeting that prolonged TTR [33], [72]. These patterns expose fragile availability and implicit trust in vendor-managed services and backup integrity. Resilience priorities include tiered segmentation and privileged-access controls, immutable/offline backups with routine restore tests, and hardened remote access (phishing-resistant MFA). Communications resilience (pre-scripted PA/SMS and cached FIDS) reduces passenger impact during portal/API outages [33], [72].

D. Comparative summary & Discussion

Despite operating at different layers (airborne navigation vs ground passenger IT), both cases show the same failure: implicit, authoritative-by-default trust lets bad/missing inputs propagate before alarms converge. Technical controls helped, but resilience depended on trust calibration and resilience-by-design (predefined degraded-mode operations, dissimilar fallbacks, time-boxed recovery). Under our Distributed Implicit Trust (DIT) framing, governed vendor dependencies plus rehearsed playbooks consistently reduced CD and recovery time (TTR).

We identify governed vendor-managed dependencies and tested degraded-mode playbooks as decisive levers for resilience, and we formalize their effects in a DIT model with propositions linking governance and practice to TTR, percent degraded services, and cascade containment. The comparison is summarized in Table III.

Implementing CD poses two practical challenges:

- Boundary definition across airline–airport–vendor–ATC layers (what counts as a step?), and
- Evidentiary sufficiency to validate each hop using time-aligned telemetry (network/app logs, SIEM/XDR) and

human/procedural records (ticketing/bridge notes, SOP deviations, documented workarounds).

TABLE III. CROSS-CASE OUTCOMES

Dimension	Case 1: GNSS Disruptions (Airborne)	Case 2: LAX DDoS (Ground)
Context / Systems	GNSS navigation (PBN), ADS-B integrity; EU Areas of Interest.	Public portals, FIDS, online/e-kiosks; edge/CDN/WAF capacity.
Threat Modality	RFI/jamming (NACp/NIC degradation); spoofing as related risk.	Hacktivist DDoS: volumetric + protocol + application-layer flooding.
Observables / Evidence	Sustained NACp=0 (>1 min); affected flights per 30-min interval; spatial clustering.	Reachability/latency errors; CDN/scrubber telemetry; user-facing outages.
Trust Assumptions Exploited	Unauthenticated GNSS signals; single-signal dependency.	Assumed availability of public services; vendor scrubbing capacity not stress-validated.
Failure Modes	Navigation integrity loss; pilot workload; ATC reliance.	Portal/FIDS unavailability; manual passenger processing; congestion.
Immediate Operational Impact	Delays/reroutes; safety preserved via redundancy and ATC support.	Delays at check-in/gates; info flow degraded; flight safety unaffected.
Fallbacks Required	IRS/DME/sensor fusion; ground-based navaids; ATC vectors.	Manual check-in/boarding; cached FIDS; PA/SMS messaging; queue runbooks.
Resilience Controls (A-A-R-A)	Anticipate: RFI monitoring; Absorb: sensor fusion; Recover: reacquire GNSS; Adapt: harden receivers, maintain navaids.	Anticipate: capacity tests; Absorb: anycast/scrubbing; Recover: failover & comms; Adapt: cached FIDS, SLAs, exercises.
Key Metrics	Share of affected flights; RFI duration; ATC assist frequency.	TTR; peak attack bandwidth vs. capacity; % degraded services; throughput impact.
Strategic Lessons	Avoid single-signal dependency; preserve multi-layer navigation.	Engineer for hostile peaks; decouple passenger comms; validate vendor capacity; drill fallbacks.

Table III. Cross-Case Outcomes—GNSS Disruption Events (Case 1 vs. Case 2). Metrics and event traces synthesized from [33], [37].

VI. CONCLUSION

Across two strata of the aviation ecosystem—airborne navigation (GNSS disruptions) and ground digital services (airport web/API denial)—we find a recurring failure mode: implicit, authoritative-by-default trust in critical inputs (PNT accuracy; service availability) lets bad or missing data reach operations before alarms converge, triggering cascading effects. Technical controls (detection, authentication, WAF/CDN) are necessary but insufficient; cascades were bounded fastest when trust calibration—multi-source validation,

challenge-response for safety-critical changes, and provenance/dependency visibility—was paired with resilience-by-design mechanisms: predefined degraded-mode operations with bounded performance, dissimilar-channel fallbacks, and time-boxed recovery objectives. Measuring CD faces two challenges: defining boundaries across airline–airport–vendor–ATC layers, and validating each hop with both telemetry (network/app/SIEM) and procedural records (tickets, runbooks). Where evidence is partial, CD can be reported as a range with uncertain vendor-managed links flagged.

We advance a trust- and resilience-centric framing for connected aviation; introduce a DIT model that formalizes governance and degraded-mode levers; present cross-case evidence of propagation paths; and deliver actionable metrics—time to detect (TTD)/TTR (or MTTD/MTTR for aggregates), recover time objective (RTO), and a domain-adapted CD, to operationalize improvement. Future work will quantify cascade reduction from specific verification gates and degraded-mode policies and embed these practices in procurement, assurance cases, and recurrent training.

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